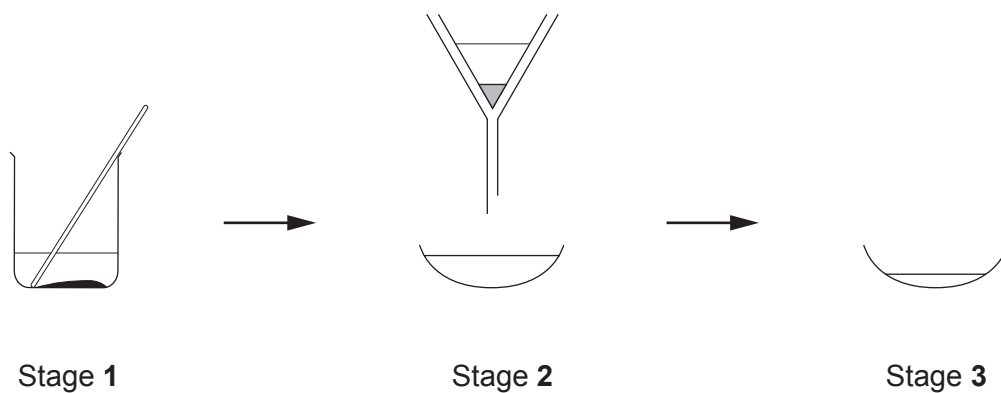


Examiner
only

6. The diagram shows the three stages used in the preparation of copper(II) sulfate crystals from copper(II) carbonate and sulfuric acid.



Describe and explain each stage of the preparation. Include an equation in your answer.

[6 QER]

6



6. Crude oil can be separated into simpler mixtures, called fractions, which contain hydrocarbon compounds with boiling points in a similar range.

(a) The table lists the properties of some fractions obtained from crude oil.

Fraction	Number of carbon atoms in fraction	Boiling point range (°C)	Colour of fraction	Flame when burning	Ease of burning
fuel oil	1-4	-170 to 20	colourless	clean	very easy
petrol	5-10	20 to 70	pale yellow	clean	easy
naphtha	8-12	70 to 120	yellow	some soot	quite easy
kerosene	10-16	120 to 240	dark yellow	smoky	quite difficult
diesel oil	15-30	240 to 350	brown	smoky	difficult

Use the information in the table to describe how the burning of the fractions depends upon the size of the molecules. [2]

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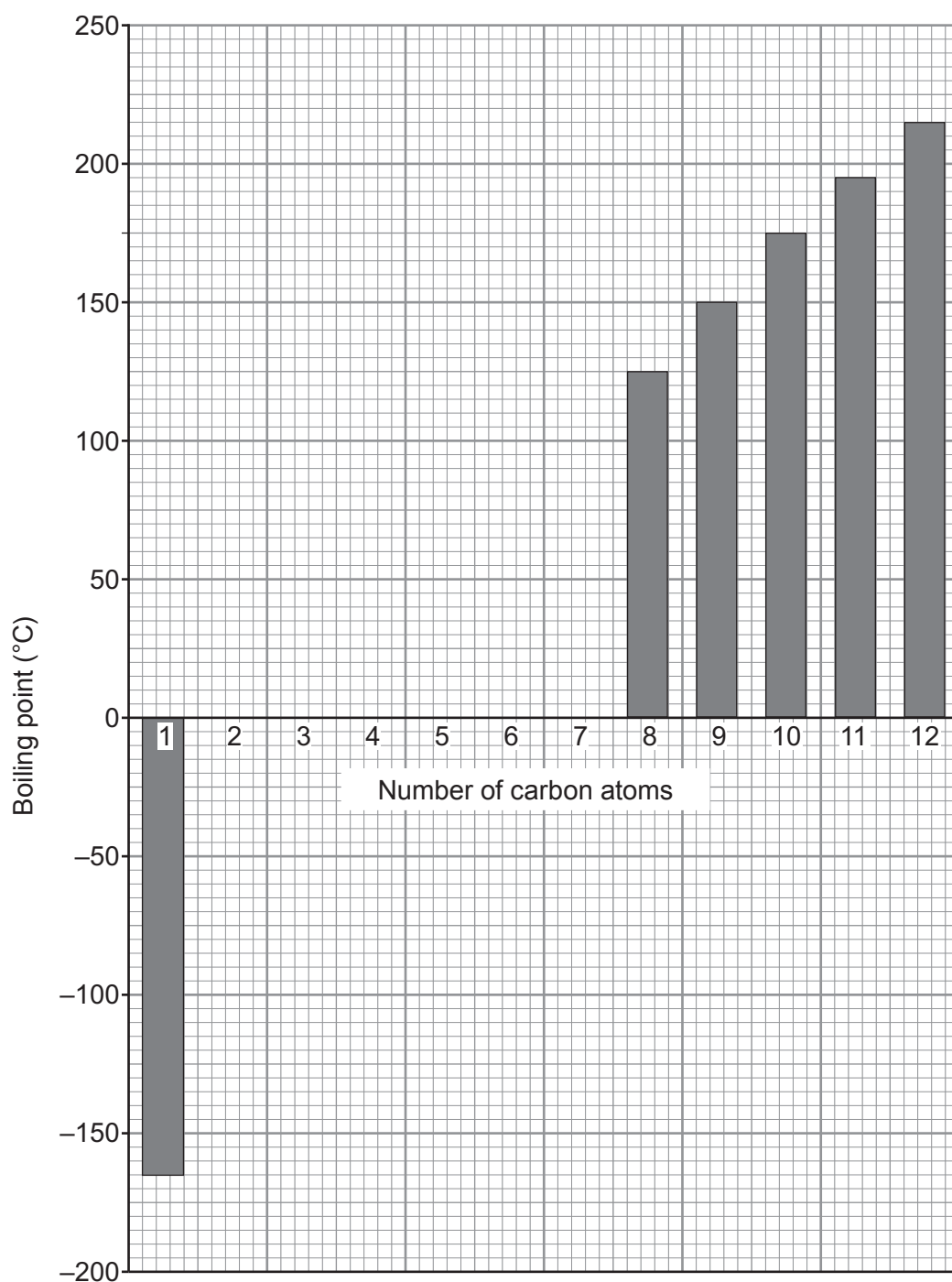
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- (b) The boiling points of hydrocarbons containing 1 to 12 carbon atoms are shown in the table below. The boiling point for the hydrocarbon with 7 carbon atoms is missing.

Number of carbon atoms	Boiling point (°C)
1	-165
2	-90
3	-40
4	10
5	35
6	70
7	
8	125
9	150
10	175
11	195
12	215



- (i) Complete the bar chart below. Some of the bars have already been drawn. [2]



- (ii) Use a ruler to draw a trend line **on the chart** and use this to estimate the boiling point of the hydrocarbon with 7 carbon atoms. [2]

Boiling point °C



(c) Many of the fractions obtained from crude oil are used as fuels.

(i) The fire triangle shows the factors necessary to start and maintain a fire.

State **one** method that could be used to safely put out a small amount of spilled petrol burning on the floor. Give the reason why your chosen method would work. [1]

Method

Reason

.....

(ii) One of the hydrocarbons in petrol is pentane, C_5H_{12} .

Complete and balance the symbol equation for the complete combustion of pentane. [2]



(iii) Hydrogen fuel cells are now used in many cars instead of petrol. The overall change inside a hydrogen fuel cell is the same as when hydrogen burns.

Explain why using hydrogen fuel cells in cars is better for the environment than petrol. [2]

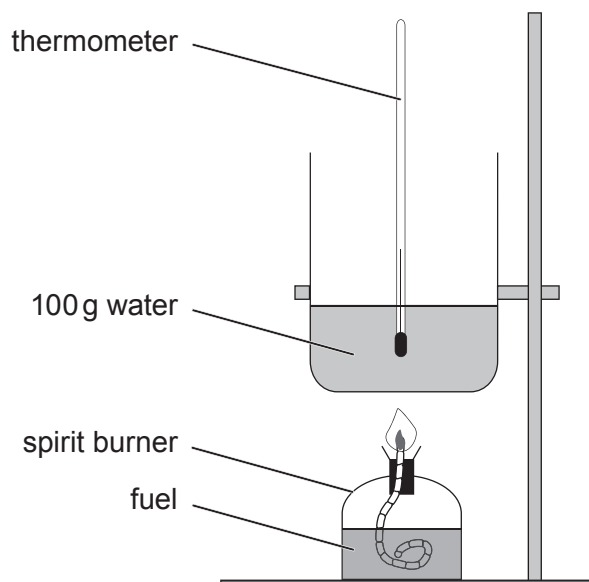
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- (d) It is possible to compare the energy released when different fuels are burned using the following apparatus.



To calculate the energy released per gram of fuel burned, the following equation is used.

$$\text{energy released per gram of fuel (J/g)} = \frac{\text{mass of water} \times 4.2 \times \text{temperature rise (}^{\circ}\text{C)}}{\text{mass of fuel used (g)}}$$

- (i) Apart from measuring the mass of water, describe **all** the measurements that would need to be taken to be able to calculate the energy released per gram of fuel burned. [2]

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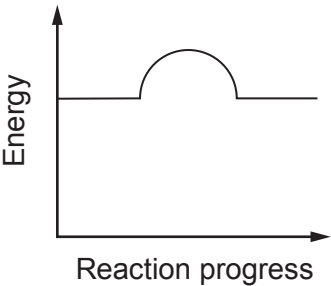
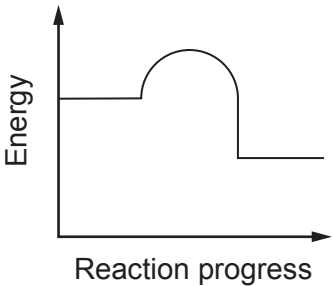
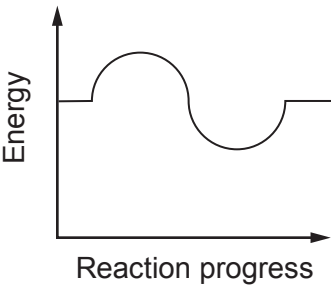
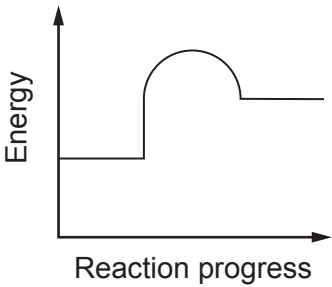
- (ii) When comparing the energy released from different fuels, 100 g of water should be used each time. State **one** other variable that should be controlled. [1]

.....

.....



(iii) Tick (✓) the energy profile diagram that represents the combustion of fuels. [1]

☐☐☐☐

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



THE PERIODIC TABLE

0

7

6

5

4

3

Group

2

1

28

1 H Hydrogen 1																	4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4	19 F Fluorine 9	20 Ne Neon 10														
23 Na Sodium 11	24 Mg Magnesium 12	16 O Oxygen 8	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18												
39 K Potassium 19	40 Ca Calcium 20	14 N Nitrogen 7	31 P Phosphorus 15	75 As Arsenic 33	84 Kr Krypton 36												
86 Rb Rubidium 37	88 Sr Strontium 38	12 C Carbon 6	28 Si Silicon 14	73 Ge Germanium 32	80 Br Bromine 35												
133 Cs Caesium 55	137 Ba Barium 56	11 B Boron 5	27 Al Aluminium 13	70 Ga Gallium 31	127 I Iodine 53												
223 Fr Francium 87	226 Ra Radium 88	59 Co Cobalt 27	56 Fe Iron 26	65 Zn Zinc 30	210 At Astatine 85												
		58 Ni Nickel 28	55 Mn Manganese 25	63.5 Cu Copper 29	222 Rn Radon 86												
		57 La Lanthanum 57	52 Cr Chromium 24	108 Ag Silver 47													
		54 Xe Xenon 54	51 V Vanadium 23	112 Cd Cadmium 48													
			48 Ti Titanium 22	110 Pb Lead 82													
			45 Sc Scandium 21	201 Hg Mercury 80													
			41 Nb Niobium 41	197 Au Gold 79													
			40 Zr Zirconium 40	195 Pt Platinum 78													
			39 Y Yttrium 39	186 Re Rhenium 75													
			37 Rb Rubidium 37	184 W Tungsten 74													
			36 Kr Krypton 36	181 Ta Tantalum 73													
			35 Br Bromine 35	179 Hf Hafnium 72													
			34 Se Selenium 34	178 Os Osmium 76													
			33 As Arsenic 33	177 Ir Iridium 77													
			32 S Sulfur 16	176 Au Gold 79													
			31 P Phosphorus 15	175 Pt Platinum 78													
			30 Zn Zinc 30	174 Au Gold 79													
			29 Cu Copper 29	173 Au Gold 79													
			28 Ni Nickel 28	172 Au Gold 79													
			27 Co Cobalt 27	171 Au Gold 79													
			26 Fe Iron 26	170 Au Gold 79													
			25 Mn Manganese 25	169 Au Gold 79													
			24 Cr Chromium 24	168 Au Gold 79													
			23 V Vanadium 23	167 Au Gold 79													
			22 Ti Titanium 22	166 Au Gold 79													
			21 Sc Scandium 21	165 Au Gold 79													
			20 Ca Calcium 20	164 Au Gold 79													
			19 K Potassium 19	163 Au Gold 79													
			18 Ar Argon 18	162 Au Gold 79													
			17 Cl Chlorine 17	161 Au Gold 79													
			16 S Sulfur 16	160 Au Gold 79													
			15 P Phosphorus 15	159 Au Gold 79													
			14 N Nitrogen 7	158 Au Gold 79													
			13 Al Aluminium 13	157 Au Gold 79													
			12 C Carbon 6	156 Au Gold 79													
			11 B Boron 5	155 Au Gold 79													
			10 Ne Neon 10	154 Au Gold 79													
			9 F Fluorine 9	153 Au Gold 79													
			8 O Oxygen 8	152 Au Gold 79													
			7 N Nitrogen 7	151 Au Gold 79													
			6 C Carbon 6	150 Au Gold 79													
			5 B Boron 5	149 Au Gold 79													
			4 He Helium 2	148 Au Gold 79													

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3430U50-1



FRIDAY, 27 MAY 2022 – MORNING

SCIENCE (Double Award)

Unit 5 – CHEMISTRY 2

FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	13	
2.	9	
3.	9	
4.	8	
5.	6	
6.	7	
7.	8	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



JUN223430U50101

6. Polymer gels are commonly used in disposable nappies.

A company that manufactures disposable nappies was investigating the effect of temperature on the mass of water the polymer gel in their nappies is able to absorb.

- (a) The results collected using water at 40 °C are given below. The initial mass of the polymer gel bead was 0.035 g.

Time (hours)	Mass of bead (g)	Mass of water absorbed by bead (g) (to 1 decimal place)
0	0.035	0.0
2	4.048	4.0
4	6.030	6.0
6	7.280	7.2
8	7.891	7.9
10	8.181	8.1
12	8.181	8.1

- (i) The percentage increase in the mass of the bead is calculated using the following equation.

percentage increase = $\frac{\text{mass of water absorbed}}{\text{initial mass of bead}} \times 100$

Calculate the percentage increase in the mass of the bead after 2 hours.
Give your answer to the nearest whole number.

[1]

Percentage increase = %

- (ii) What property of polymer gels does the figure calculated in part (i) demonstrate?

[1]

.....

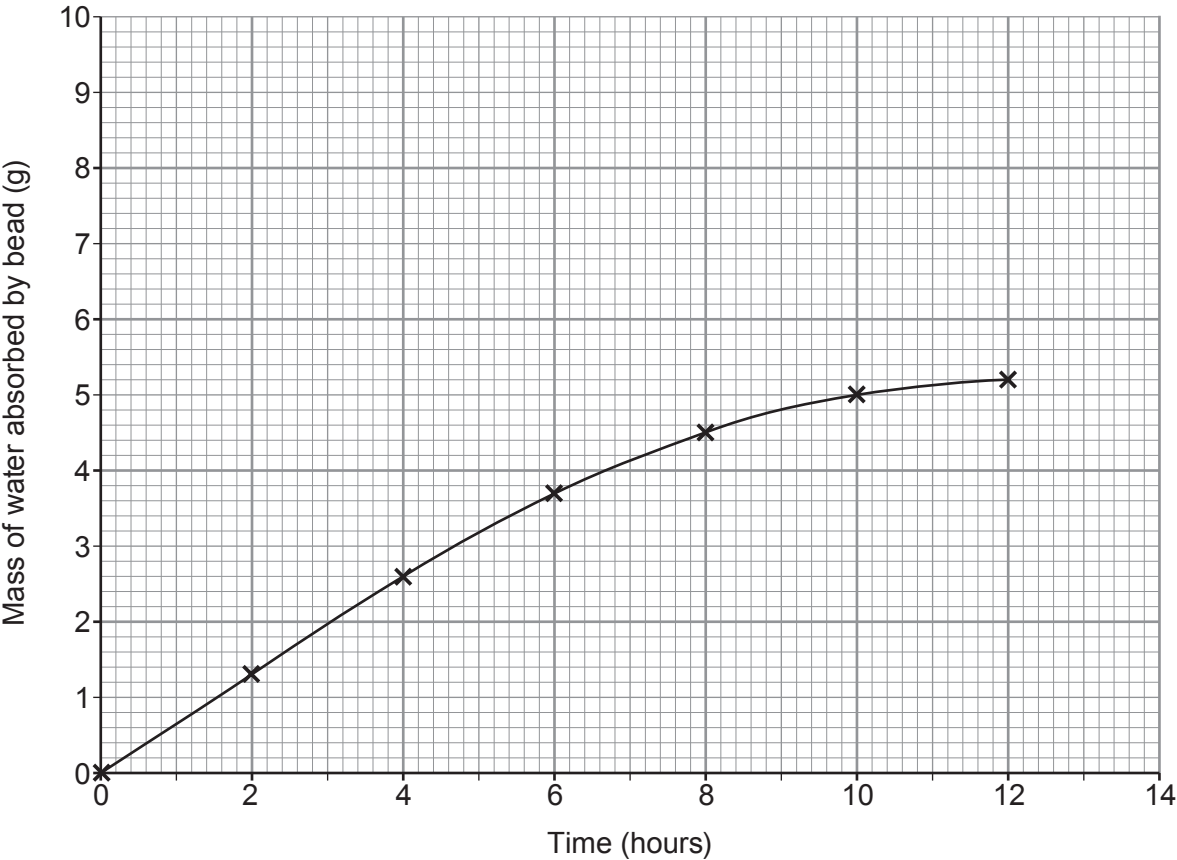


Examiner
only

- (b) (i) On the grid below, plot the results using water at 40 °C and draw a suitable line. Use the mass of water absorbed by the bead to 1 decimal place.

The results using water at 10 °C have already been plotted.

[3]



- (ii) Give **two** differences between the absorbing properties of the bead using water at 10 °C and at 40 °C.

[2]

Difference 1

.....

Difference 2

.....

7



Examiner
only

6. The table shows some properties of aluminium, copper and titanium.

Metal	Melting point	Electrical conductivity	Thermal conductivity	Density	Resistance to corrosion
aluminium	high	good	good	low	good
copper	high	good	good	high	good
titanium	high	good	good	low	good

Describe how **these** properties are linked to the uses of aluminium, copper and titanium.

You should **not** give more than two uses for any one metal. [6 QER]

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